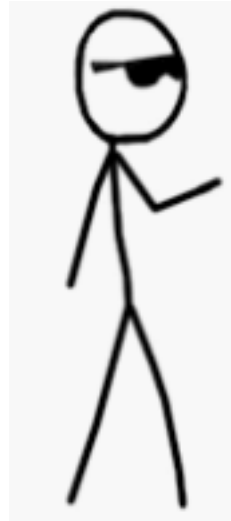

S.M Modaghegh
Vision Workshop
Gharjistan
AI
MACHINE
LEARNING



Human can learn from past experience
and make decision of its own

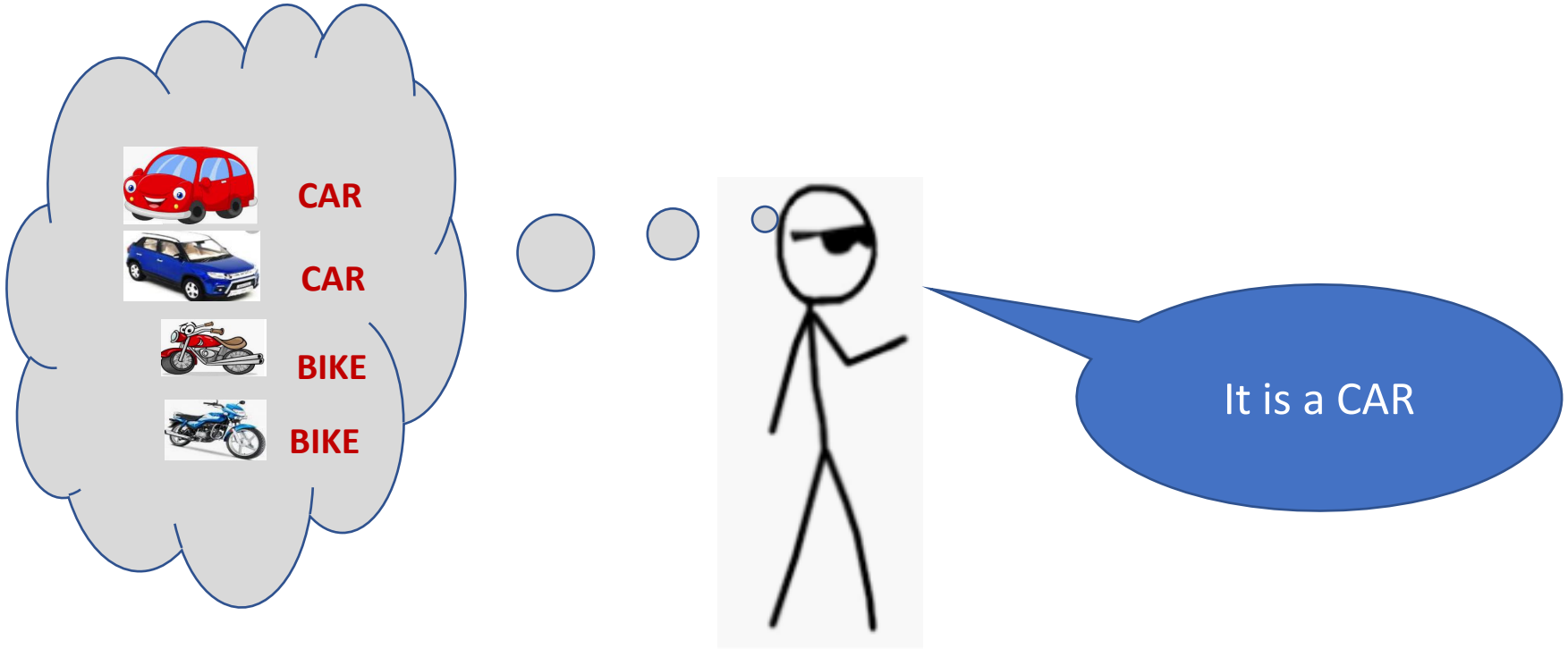


What is this object?





What is this object?



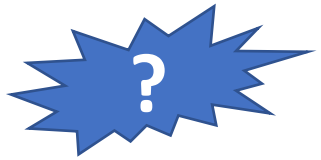
Let us ask the same question to him

What is this object?





Let us ask the same question to him



What is this object?







[But, he is a human being. He can observe and learn]

Let us make him learn



show him









Let us make him learn



show him



CAR



CAR



BIKE



BIKE

Let us ask the same question now

What is this object?







CAR



CAR



BIKE



BIKE

Past experience

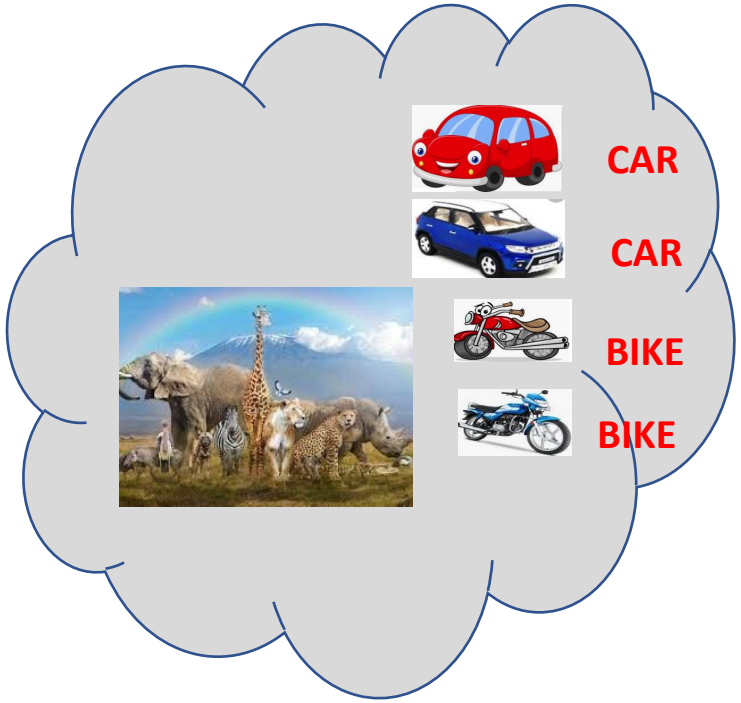
Let us ask the same question now

CAR

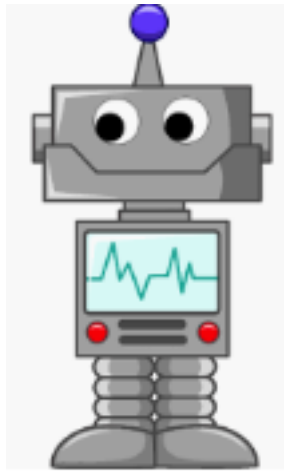
What is this object?







What about a Machine ?



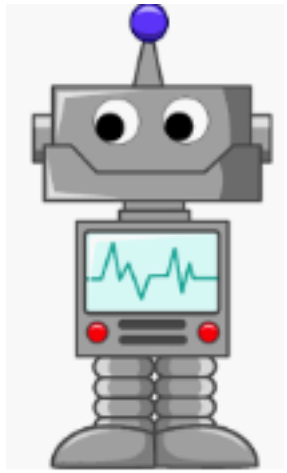
Machines follow instructions

[It can not take decision of its own]

What about a Machine ?

We can ask a machine

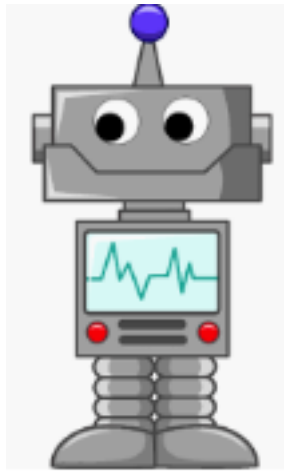
- To perform an arithmetic operations such as
  - Addition
  - Multiplication
  - Division



Machines follow instructions

What about a Machine ?

- Comparison
- Print
- Plotting a chart



Machines follow instructions

What is Machine Learning?

[We want a machine to act like a human]

What is Machine Learning?





[to identify this object.]

What is Machine Learning?





Price in 2025?

[predict the price in future]

What is Machine Learning?

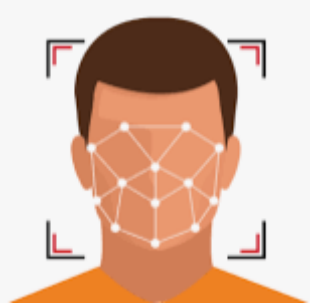


I ~~made~~ met him yesterday

[Natural Language understand, and correct grammar]

What is Machine Learning?





recognize face

[Recognize Faces]

What is Machine Learning?



[What do we do?

Just like, what we did to human,

we need to provide experience to the machine.

]

What is Machine Learning?



+



Dataset

[
This what we called as Data or Training dataset

So, we first need to provide training dataset to the machine
]

What is Machine Learning?

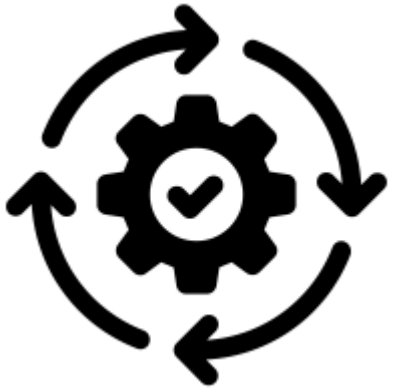


+



Dataset

+



[Then, devise algorithms and execute programs on the data

With respect to the underlying target tasks]

What is Machine Learning?

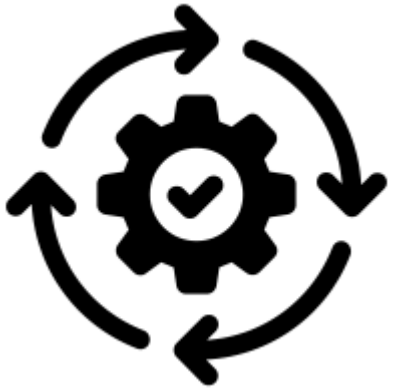


+



Dataset

+



+



[Then, using the programs, Identify required rules]

What is Machine Learning?

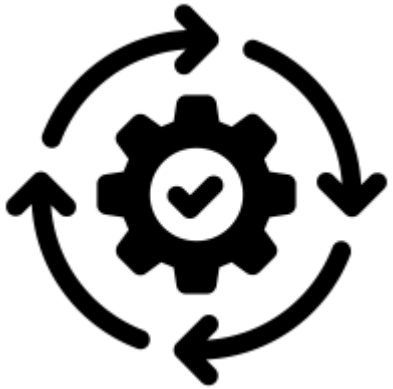


+



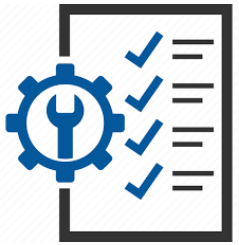
Dataset

+



+





[extract required patterns]

What is Machine Learning?

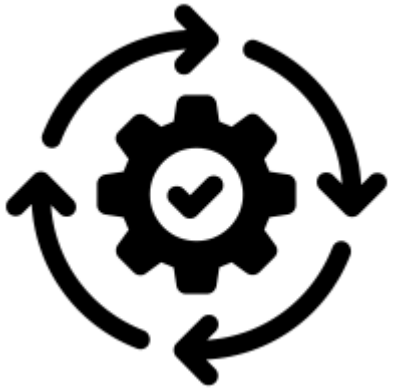


+



Dataset

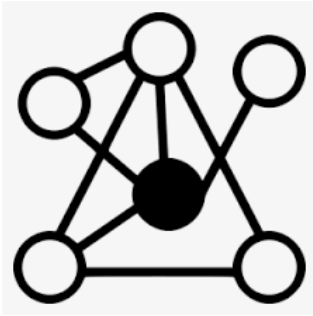
+



+

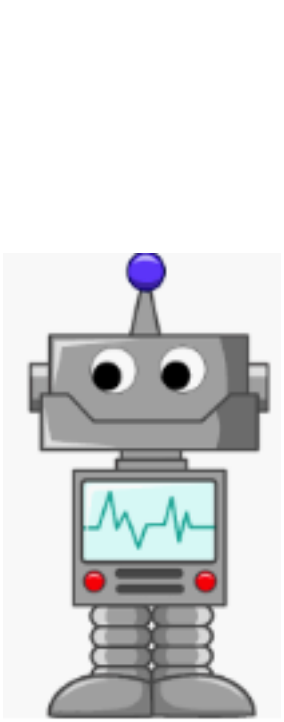






[Identify relations]

What is Machine Learning?

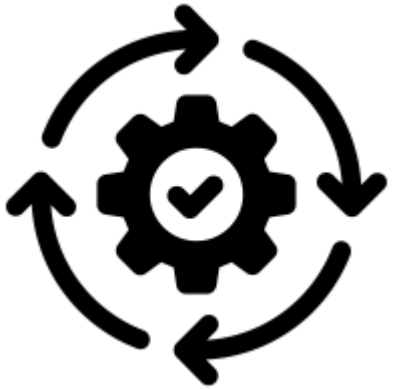


+



Dataset

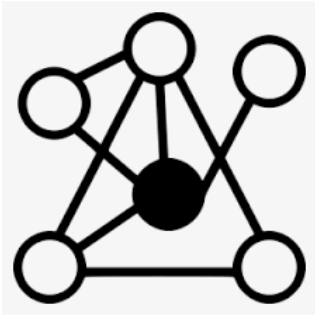
+



+







=



[So that machine can derive inferences from the data]

Machine Learning Paradigms

- Supervised
- Unsupervised Learning
- Reinforcement learning

[We as human being solve various types of problem in our day-to-day life, <pause> Various decisions need to be taken.
Depending on the nature of the problem, machine learning tasks can be broadly divided in]

What is Supervised Learning?









Samples

CAR

CAR

BIKE

BIKE

Labels

+

= **Training Dataset**

[In supervised learning, we need some thing called a Labelled Training Dataset]

What is Supervised Learning?



CAR



CAR

+



BIKE



BIKE

Samples Labels

= **Training Dataset**

$f($  , $)=$

[Given a labelled dataset, the task is to devise a function which takes the dataset, and a new sample, and produces an output value.]

What is Supervised Learning?



CAR



CAR

+



BIKE



BIKE

Samples Labels

= **Training Dataset**

$f($  ,  $)=$

[Given a labelled dataset, the task is to devise a function which takes the dataset, and a new sample, and produces an output value.]

What is Supervised Learning?


CAR


CAR

\+


BIKE


BIKE

Samples Labels

= **Training Dataset**

$f($  ,  $)$ = CAR

[Given a labelled dataset, the task is to devise a function which takes the dataset, and a new sample, and produces an output value.]

What is Supervised Learning?






CAR

CAR

BIKE

BIKE

Samples

Labels

Training Dataset

Classification

$f($  ,  $)=$ CAR

[If the possible output values of the function are predefined and discrete/categorical, it is called Classification

What is Supervised Learning?



CAR



CAR

\+



BIKE



BIKE

Samples Labels

= **Training Dataset**

Classification

$f($  $)=$ CAR

[Predefined classes means, it will produce output only from the labels defined in the dataset. For example, even if we input a bus, it will produce either CAR or BIKE]

Classifier



Elephant



Tiger

Dataset

Elephant

Classifier



Identify the Animal ?

Regression



Dataset

Regression

$f($  ,  $) = 20500.50$

[If the possible output values of the function are continuous real values, then it is called Regression

[
The classification and Regression problems are supervised, because the decision depends on the characteristics of the ground truth labels or values present in the dataset, which we define as experience
]

What is Unsupervised Learning


~~CAR~~


~~CAR~~


~~BIKE~~


~~BIKE~~

Dataset

[In the unsupervised learning, we do not need to know the labels or Ground truth values]

What is Unsupervised Learning



Dataset





Clustering

[The task is to identify the patterns like group the similar objects together]

What is Unsupervised Learning









Dataset









Association Rules Mining

[Association rules like]

More Example Unsupervised Learning



Dataset

More Example Unsupervised Learning



Dataset



More Example Unsupervised Learning







What is Reinforcement Learning

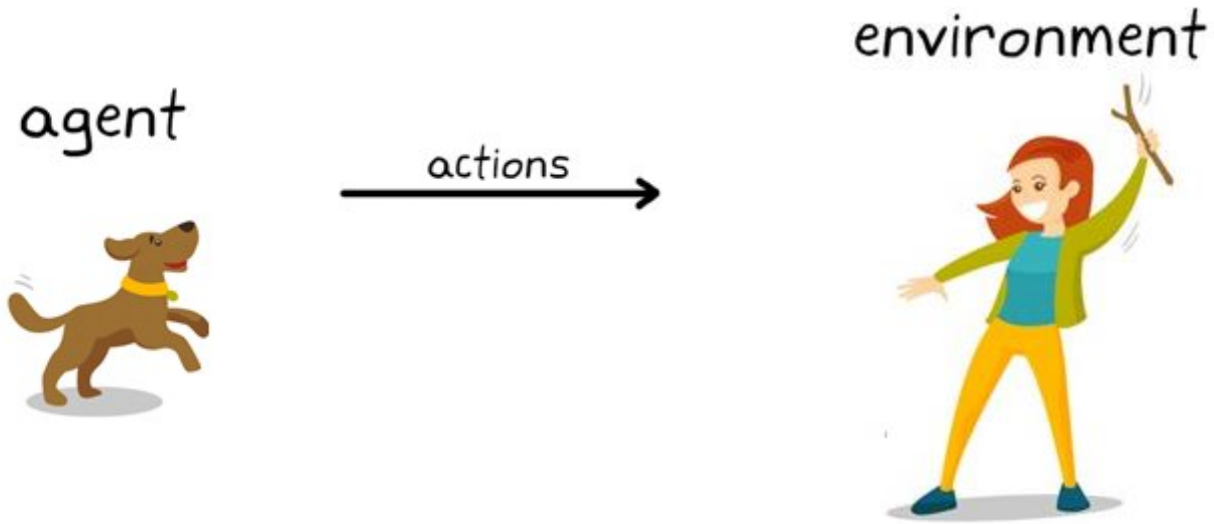
[It is also known as learning from trials and errors]

What is Reinforcement Learning

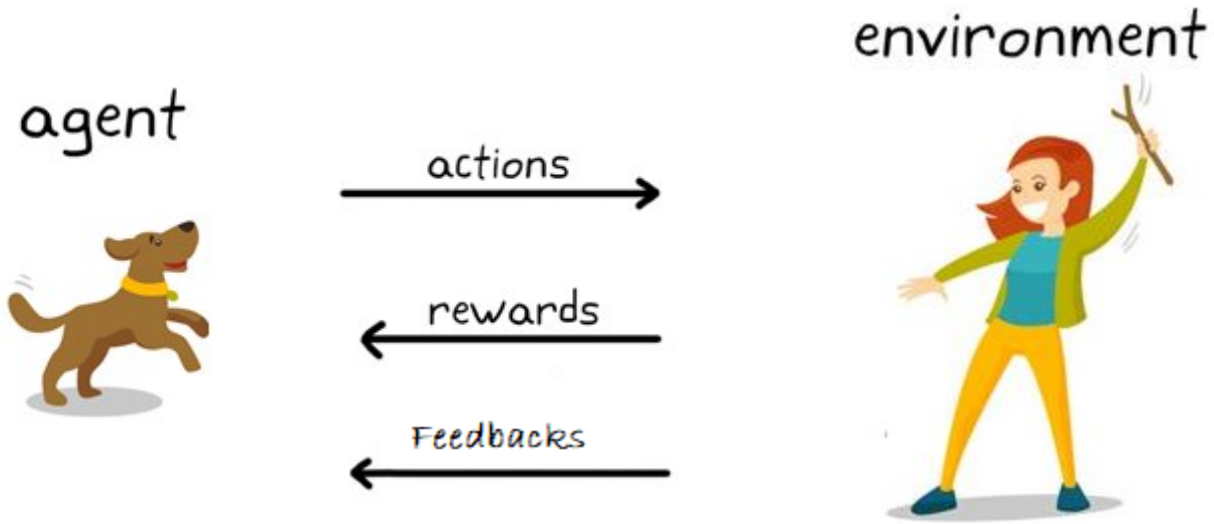




What is Reinforcement Learning



What is Reinforcement Learning



Another Example



Agent



Task



Environment

Reinforcement Learning









Punishment

Reinforcement Learning









Reward

Reinforcement Learning







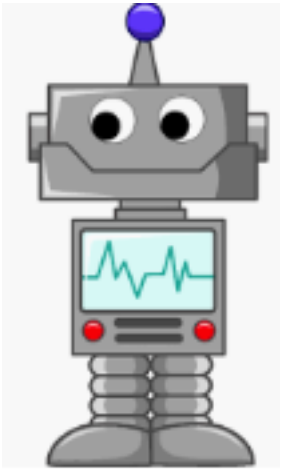


Reward

Baby Learn from the Trials and Errors

Reinforcement Learning

Teach a machine to identify vehicle types











Represent the sample







#Wheel	Height	Weight	Color

Represent the sample





#Wheel Height Weight Color

Identify the features which can represent the objects

$$F = \{f_1 f_2 f_3 \dots f_k\}$$

Feature set={ #Wheel Height Weight Color }

Represent the sample





#Wheel Height Weight Color

Identify the features which can represent the objects

$$F = \{f_1 f_2 f_3 \dots f_k\}$$

For every sample, assign value to corresponding feature

$$v_i = \{w_{i1} w_{i2} w_{i3} \dots w_{ik}\}$$

where w_{ij} is the value assigned for the feature f_j

Represent the sample





#Wheel	Height	Weight	Color
4	6	500	Red
4	5.5	600	Blue
4	5	550	Yellow
2	3	200	Red
2	3.5	150	blue
2	4	250	Yellow

For every object, assign value to corresponding feature

$$v_i = \{w_{i1} w_{i2} w_{i3} \dots w_{ik}\}$$

where w_{ij} is the value assigned for the feature f_j

Vector Space Model





#Wheel	Height	Weight	Color
4	6	500	Red
4	5.5	600	Blue
4	5	550	Yellow
2	3	200	Red
2	3.5	150	blue
2	4	250	Yellow

Features Vectors

This form of representation is called Vector Space Model

Are all features useful?





#Wheel	Height	Weight	Color
4	6	500	Red
4	5.5	600	Blue
4	5	550	Yellow
2	3	200	Red
2	3.5	150	blue
2	4	250	Yellow

Features

Features Vectors

Good Features

- #Wheel
- Height
- Weight

Bad Feature

- Colour

Let us consider single feature





#Wheel	Class Label
4	CAR
4	CAR
4	CAR
2	BIKE
2	BIKE
2	BIKE

Training Dataset

Feature vector with Class label

Given the #Wheel, identify the vehicle



	#Wheel	Class Label
	4	CAR
	4	CAR
	4	CAR
	2	BIKE
	2	BIKE
	2	BIKE
	2	?

Let us estimate







#Wheel	Class Label
4	CAR
4	CAR
4	CAR
2	BIKE
2	BIKE
2	BIKE

Pr(Vehicle type| #Wheel) = ?

Let us estimate the probability (type|#wheel)





#Wheel	Class Label
4	CAR
4	CAR
4	CAR
2	BIKE
2	BIKE
2	BIKE

Pr(CAR| 4) = 100%

Pr(BIKE| 4) = 0%

Pr(CAR| 2) = 0%

Pr(BIKE| 2) = 100%

Ask the question now



	#Wheel	Class Label
	4	CAR
	4	CAR
	4	CAR
	2	BIKE
	2	BIKE
	2	BIKE
	2	

Pr(CAR| 4) = 100%

Pr(BIKE| 4) = 0%

Pr(CAR| 2) = 0%

Pr(BIKE| 2) = 100%



{2} ?

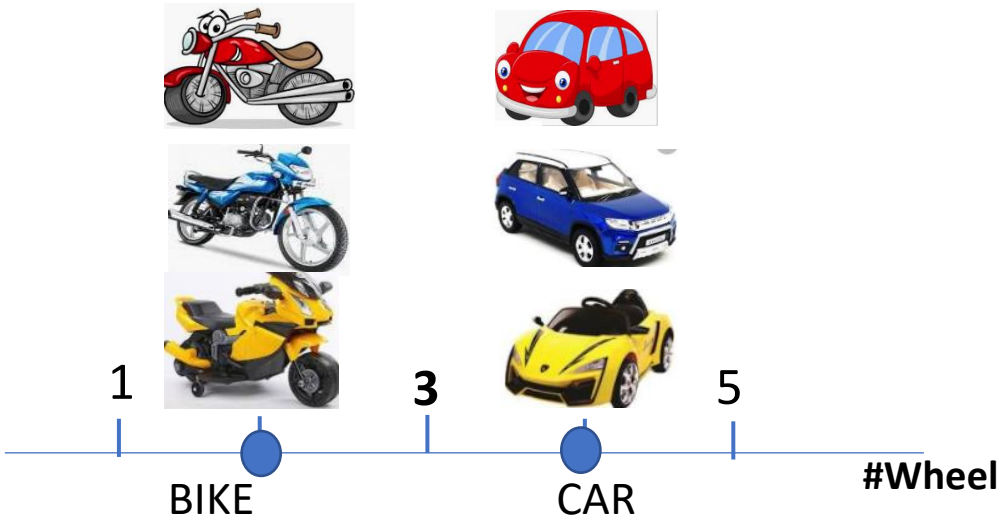


There are multiple ways





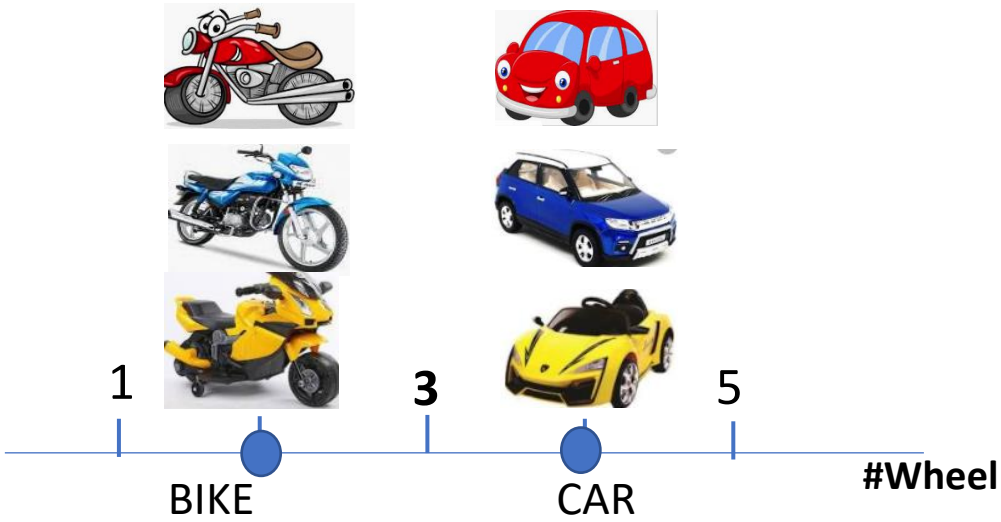
#Wheel	Class Label
4	CAR
4	CAR
4	CAR
2	BIKE
2	BIKE
2	BIKE



There are multiple ways



	#Wheel	Class Label
	4	CAR
	4	CAR
	4	CAR
	2	BIKE
	2	BIKE
	2	BIKE
	2	



Classifier
If #Wheel < 3, then it is BIKE

BIKE

If selected feature is not sufficient







#Wheel	Class Label
4	CAR
4	CAR
4	CAR
2	BIKE
2	BIKE
2	BIKE
4	BIKE
2	CAR

Pr(CAR| 4) = 75%

Pr(BIKE| 4) = 25%

Pr(CAR| 2) = 25%

Pr(BIKE| 2) = 75%



2 ?

If selected feature is not sufficient



	#Wheel	Class Label
	4	CAR
	4	CAR
	4	CAR
	2	BIKE
	2	BIKE
	2	BIKE
	4	BIKE
	2	CAR

Pr(CAR| 4) = 75%

Pr(BIKE| 4) = 25%

Pr(CAR| 2) = 25%

Pr(BIKE| 2) = 75%



2 **BIKE**

Pr(BIKE|2) > Pr(CAR|2) => BIKE

More Features







#Wheel	Height	Class Label
4	H	CAR
4	H	CAR
4	H	CAR
2	L	BIKE
2	L	BIKE
2	L	BIKE
4	L	BIKE
2	H	CAR

H: High, height >= 5

L: Low, height < 5

Estimate the probabilities, and ask the same question



	#Wheel	Height	Class Label
	4	H	CAR
	4	H	CAR
	4	H	CAR
	2	L	BIKE
	2	L	BIKE
	2	L	BIKE
	4	L	BIKE
	2	H	CAR

Pr(CAR| 4,H) = 100%

Pr(BIKE| 4,L) = 100%

Pr(CAR| 2,H) = 100%

Pr(BIKE| 2,L) = 100%

Pr(CAR| 4,L) = 0%

Pr(BIKE|4,H) = 0%

Pr(CAR| 2,L) = 0%

Pr(BIKE| 2,H) = 0%



{2 H}

?

Estimate the probabilities, and ask the same question



	#Wheel	Height	Class Label
	4	H	CAR
	4	H	CAR
	4	H	CAR
	2	L	BIKE
	2	L	BIKE
	2	L	BIKE
	4	L	BIKE
	2	H	CAR

Pr(CAR| 4,H) = 100%

Pr(BIKE| 4,L) = 100%

Pr(CAR| 2,H) = 100%

Pr(BIKE| 2,L) = 100%

Pr(CAR| 4,L) = 0%

Pr(BIKE|4,H) = 0%

Pr(CAR| 2,L) = 0%

Pr(BIKE| 2,H) = 0%



{2 H}

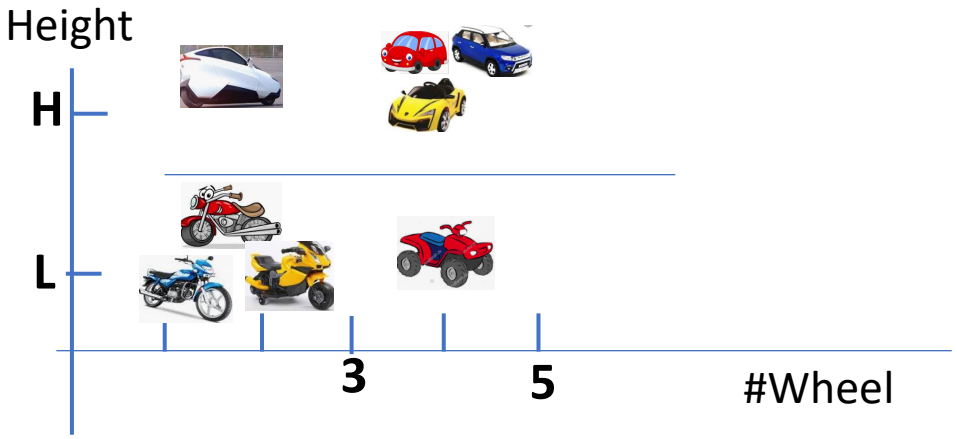
CAR

Multiple ways





#Wheel	Height	Class Label
4	H	CAR
4	H	CAR
4	H	CAR
2	L	BIKE
2	L	BIKE
2	L	BIKE
4	L	BIKE
2	H	CAR

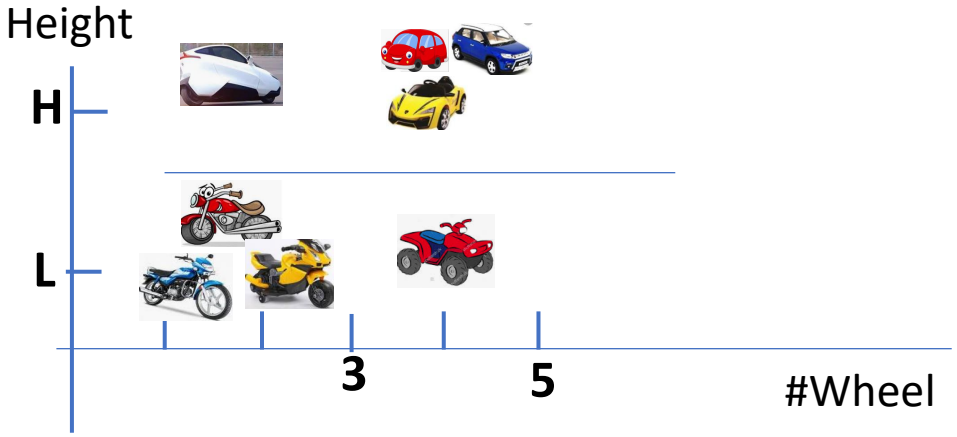


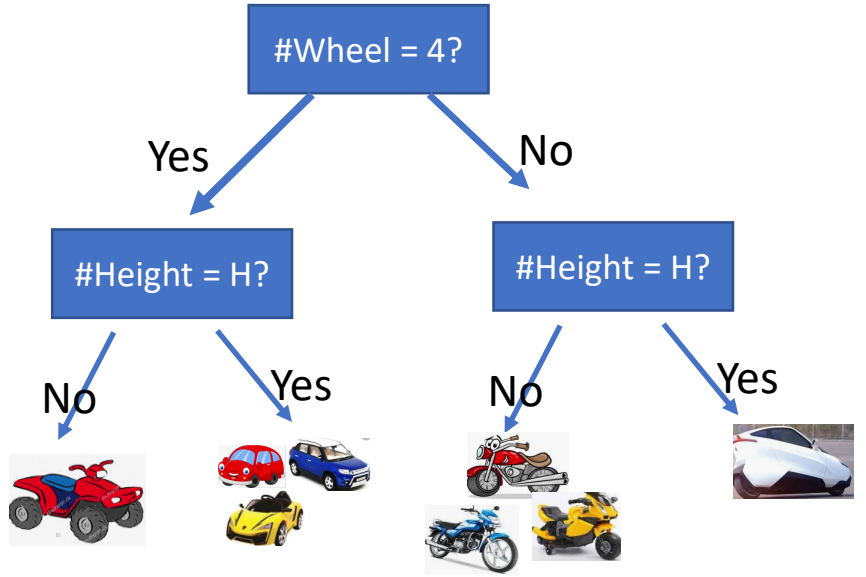
Multiple ways





#Wheel	Height	Class Label
4	H	CAR
4	H	CAR
4	H	CAR
2	L	BIKE
2	L	BIKE
2	L	BIKE
4	L	BIKE
2	H	CAR

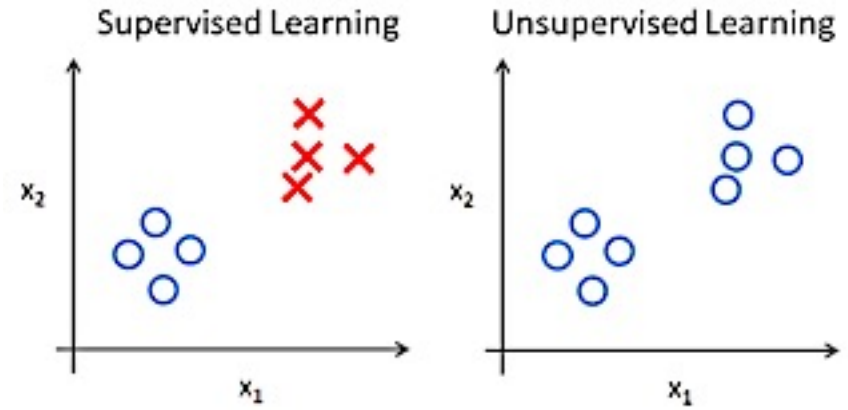




Supervised learning

- By far the most common case.
- It consists of learning to map input data to known targets, given a set of examples (often annotated by humans).
- Main categories: classification and regression.





Classification vs. Regression



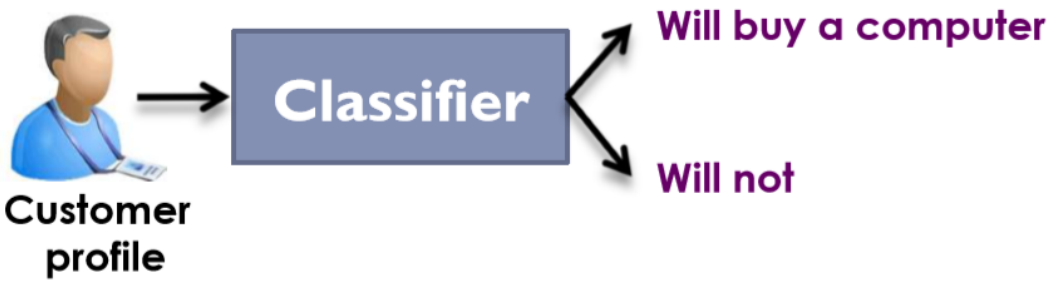
- Predicts categorical class labels (discrete or nominal).
- Example: What is the weather status tomorrow (rainy, cloudy, sunny)?



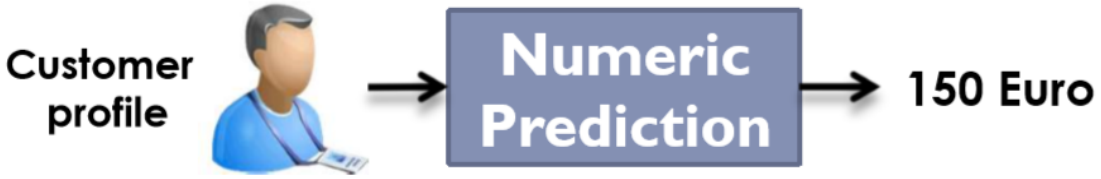
- Models continuous-valued functions.
- Example: What is the temperature going to be tomorrow?

Example

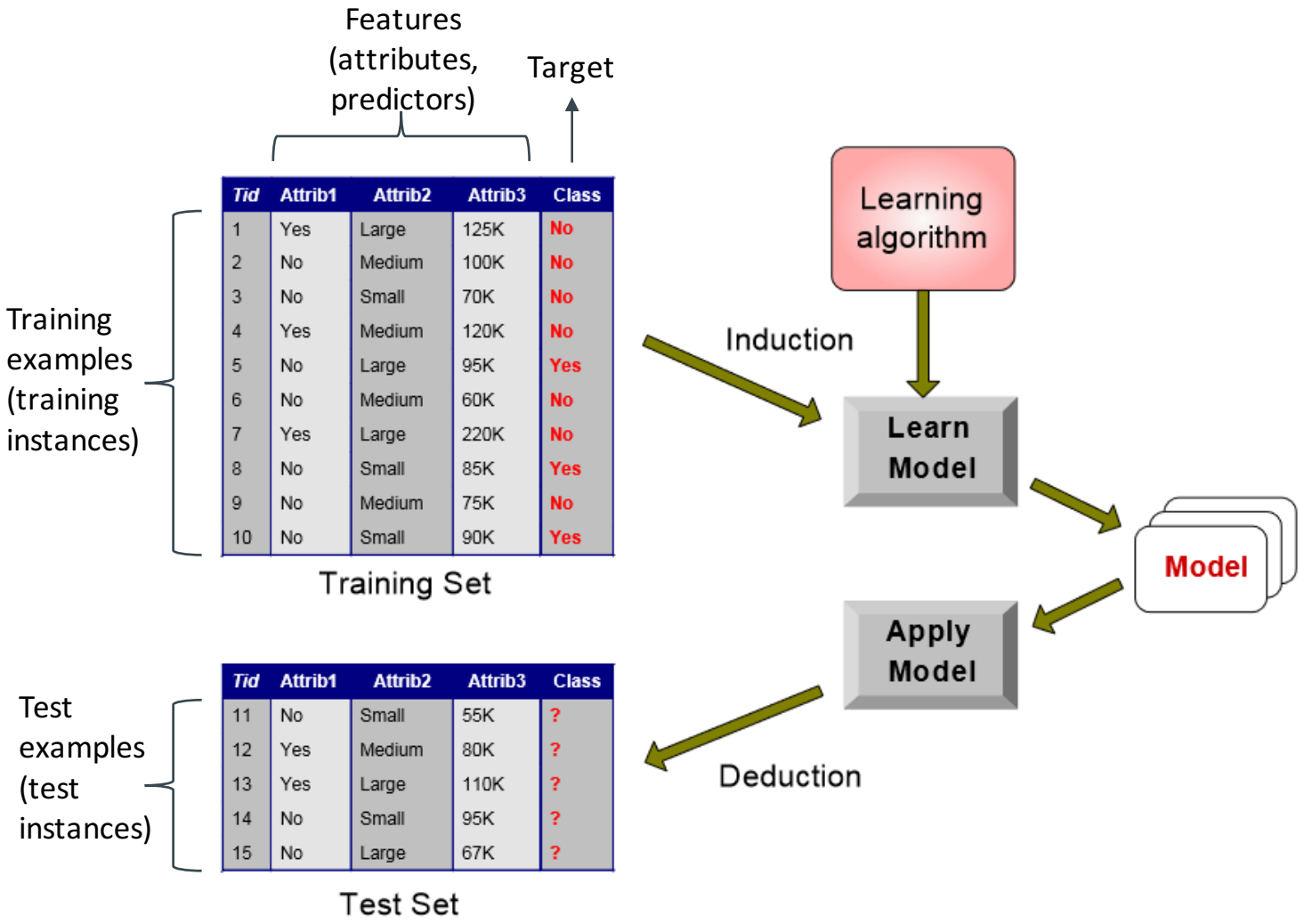
A marketing manager would like to know whether a given customer will buy a product or not



A marketing manager would like to predict how much a given customer will spend during a sale

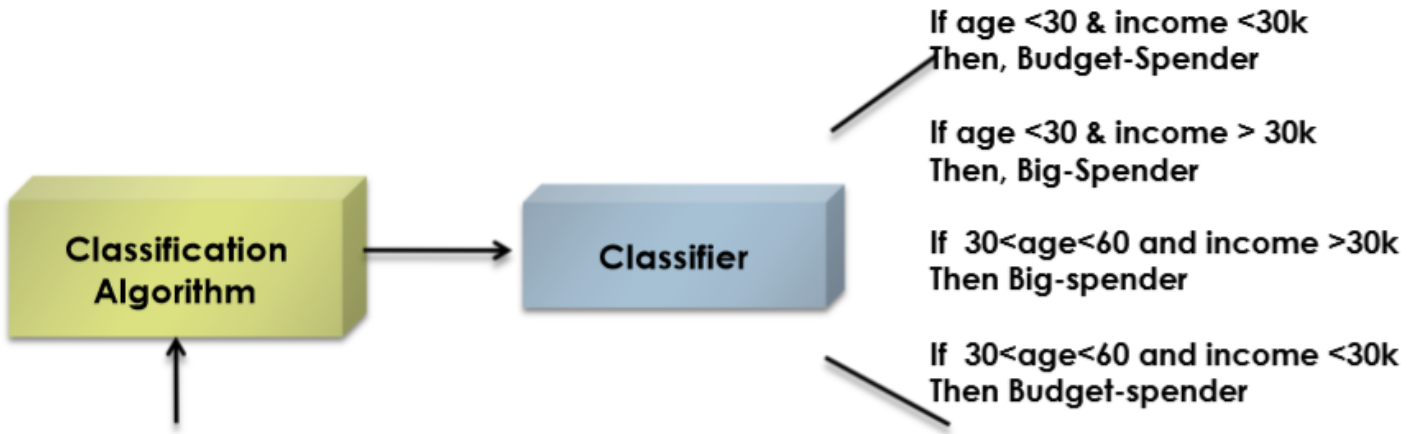


Classification



The model is represented as classification rules, decision trees, mathematical formulas, etc.

Training phase



Training data

Age	Income	Class label
27	28K	Budget-Spender
35	36K	Big-Spender
38	28K	Budget-Spender
65	45K	Budget-Spender
20	18k	Budget-Spender
75	40k	Budget-Spender
28	50k	Big-Spender
40	60k	Big-Spender
60	65k	Big-Spender

Evaluation and usage phases

1-Test the classifier

Test Data

Age	Income	Class label
27	28K	Budget-Spenders
25	36K	Big-Spenders
70	45K	Budget-Spenders
40	35k	Big-Spender



2-If acceptable accuracy

Unlabeled data

Age	Income
18	28K
37	40K
60	45K
40	36k



Classified data

Age	Income	Class label
18	28K	Budget-Spenders
37	40K	Big-Spenders
60	45K	Budget-Spenders
40	36k	Budget-Spenders

Binary vs. Multiclass classification



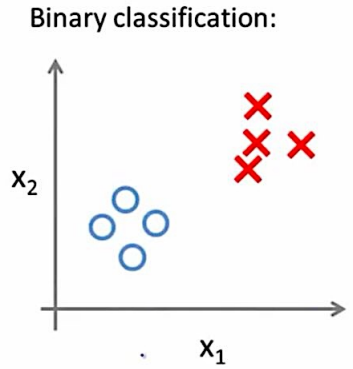
Binary classification

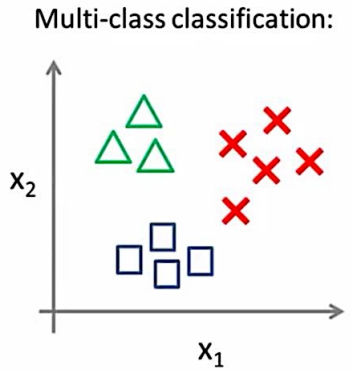
Classification tasks with only two classes, typically denoted by {+,−}, {+1,−1}, or {Pos, Neg}.
Example: email spam detection, (pos/neg) sentiment analysis.



Multiclass classification

Classification tasks with more than two classes.
Example: email topic detection, (pos/null/neg) sentiment analysis.



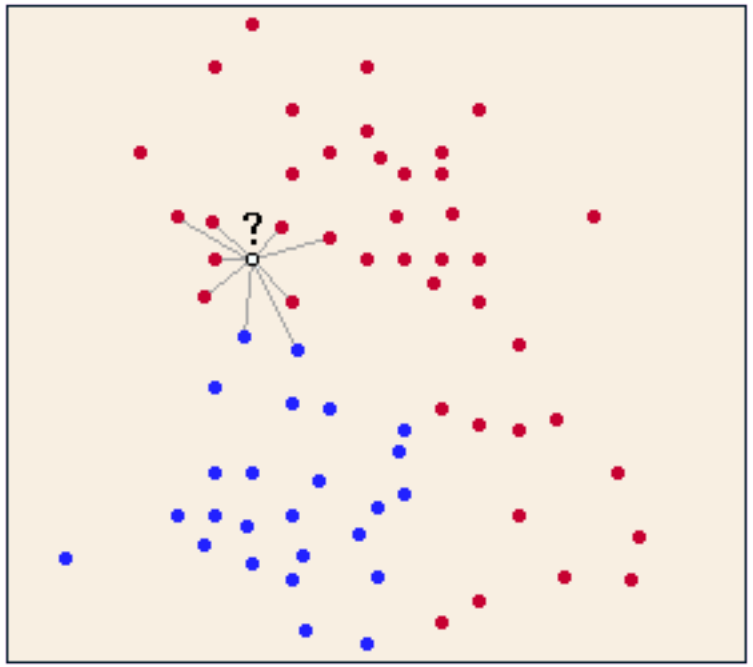


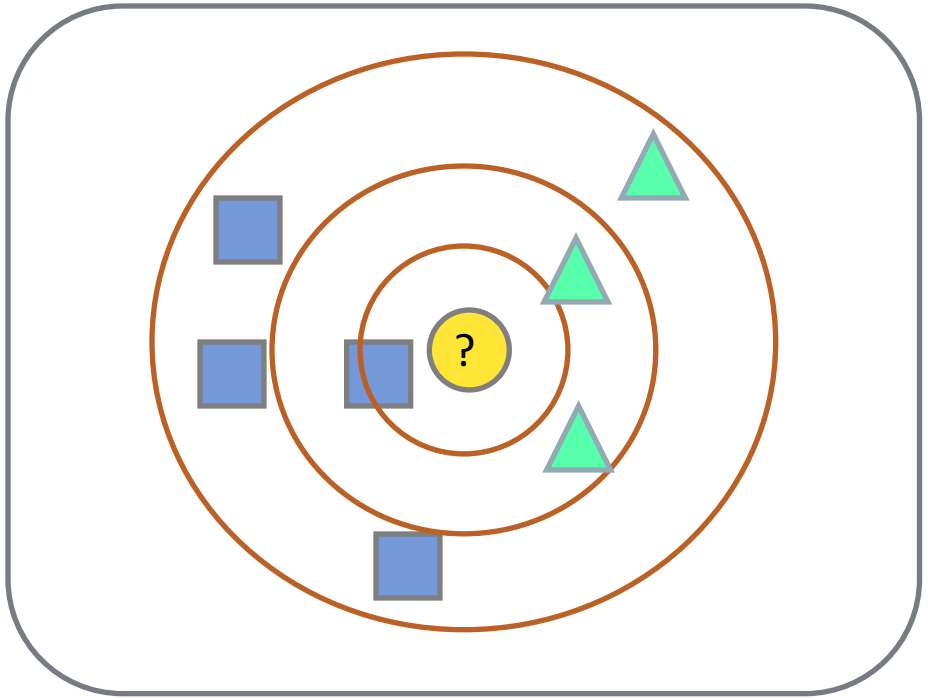
Classification algorithms

- Decision tree
- Bayesian methods (Naïve Bayes and Bayesian networks)
- K-Nearest Neighbors (KNN)
- Logistic regression
- Support Vector Machines (SVM)
- Artificial neural networks and Deep learning
- Ensemble learning
- And many more…

K-Nearest Neighbors (KNN)

KNN





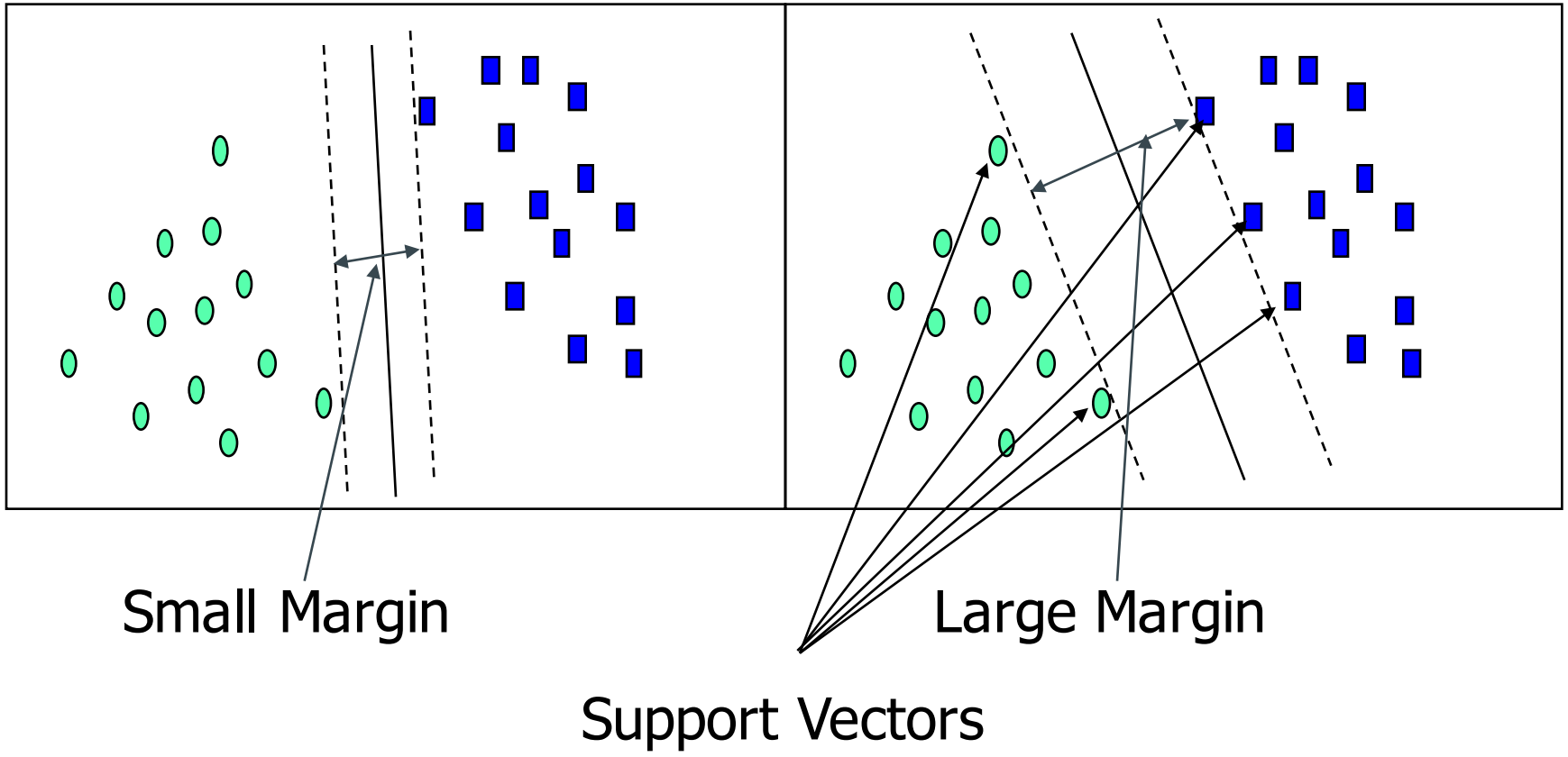
- k = 1:
  - Belongs to square class
- k = 3:
  - Belongs to triangle class
- k = 7:
  - Belongs to square class

KNN

- Among the simplest of all data mining algorithms
- Among the top 10 data mining algorithms
  - Wu, Xindong, et al. "Top 10 algorithms in data mining." *Knowledge and information systems* 14.1 (2008): 1-37.
- Requires 3 things:
  - Feature space (training data)
  - Distance metric
  - The value of *k*

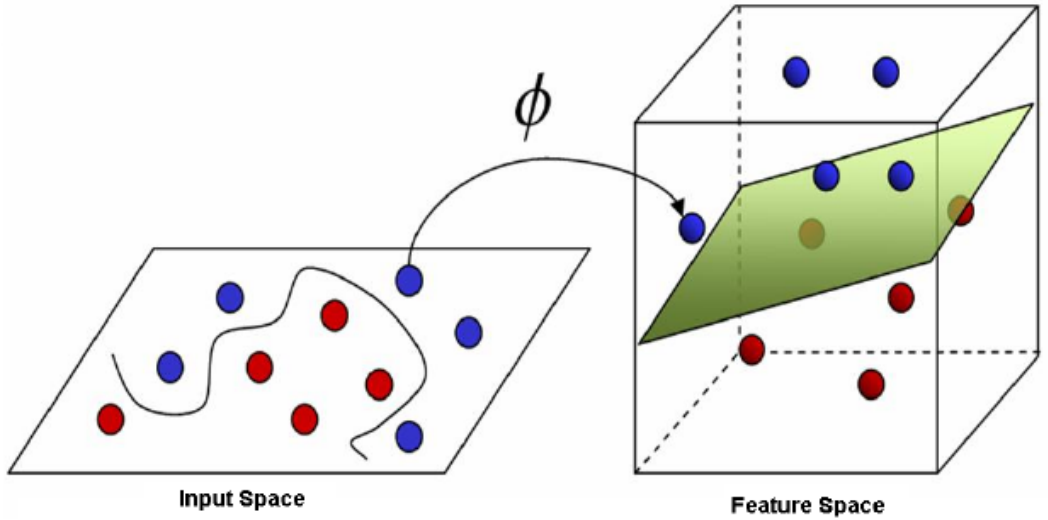
Support Vector Machine (SVM)

SVM



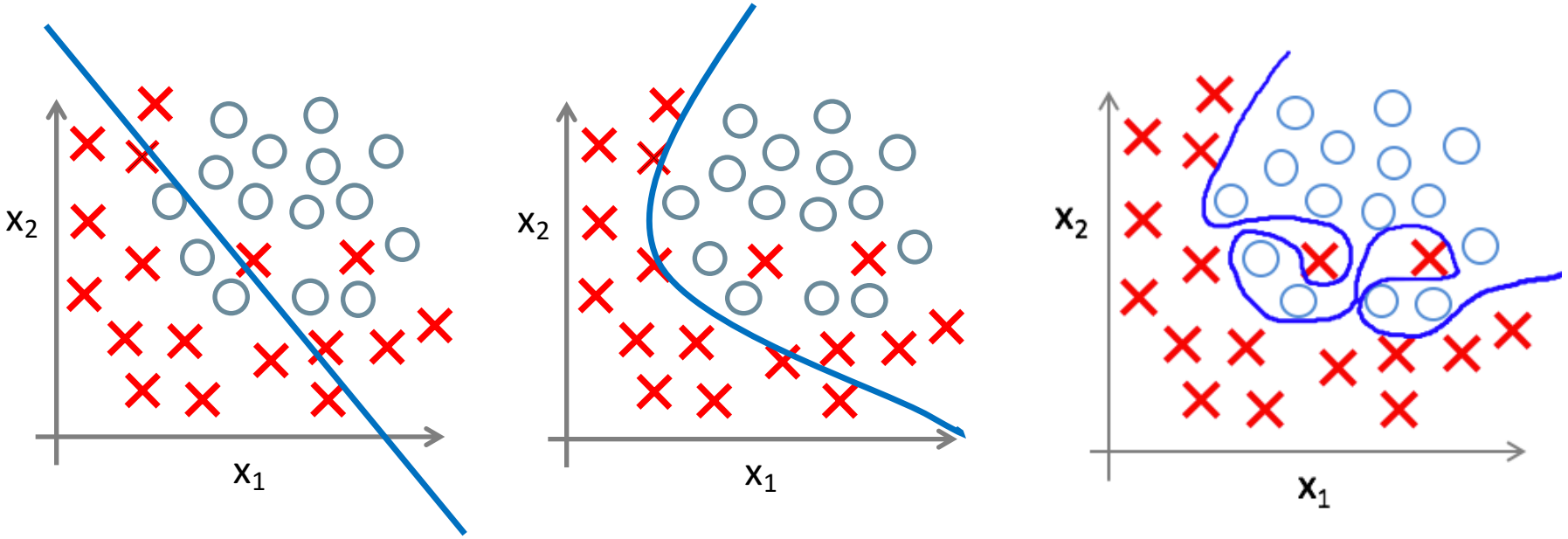
SVM

- With an appropriate nonlinear mapping to a sufficiently high dimension (kernel trick), data from two classes can always be separated by a hyperplane.
- With the new dimension, SVM searches for the optimal linear decision boundary.



http://thecaffeinedev.com/2-support-vector-machine-learning-math-behind-part2/

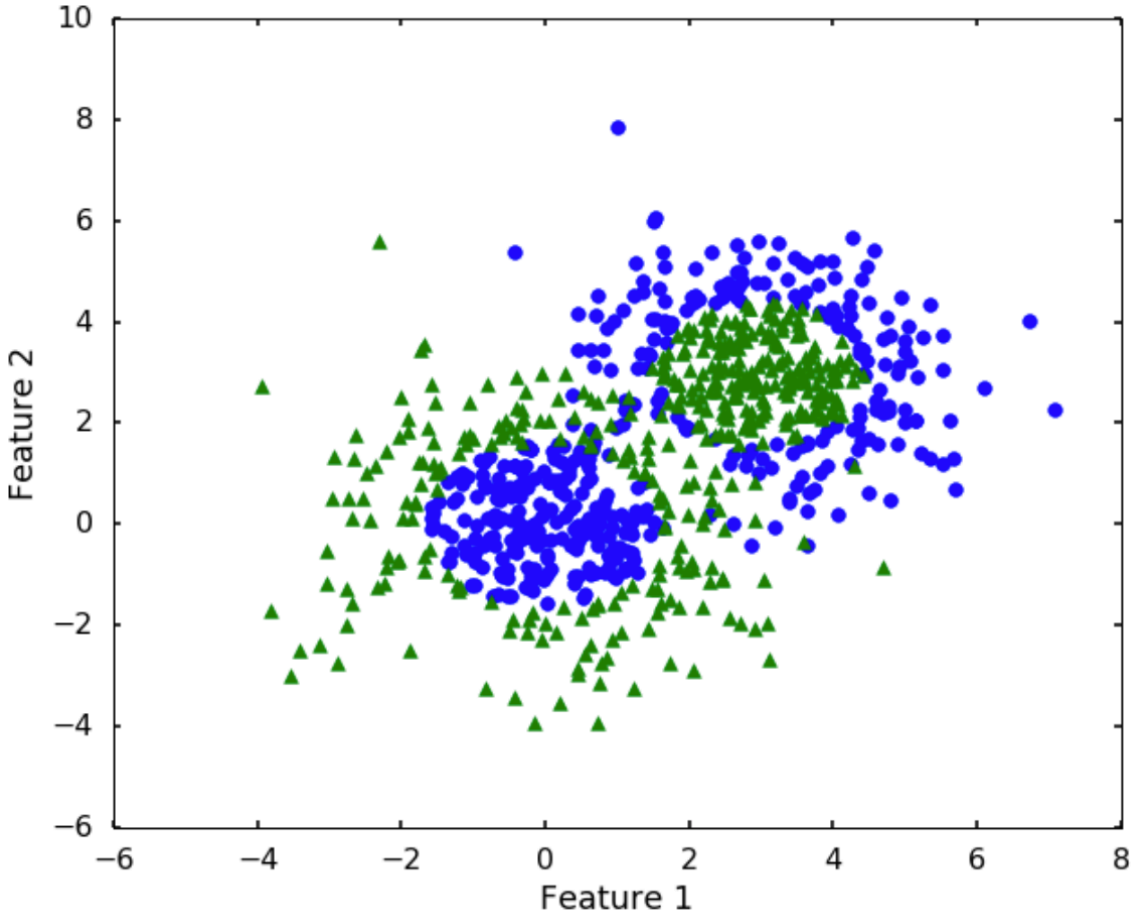
SVM



Underfit OK Overfit

In sklearn, this is controlled by C parameter of SVM and gamma parameter of rbf kernel.

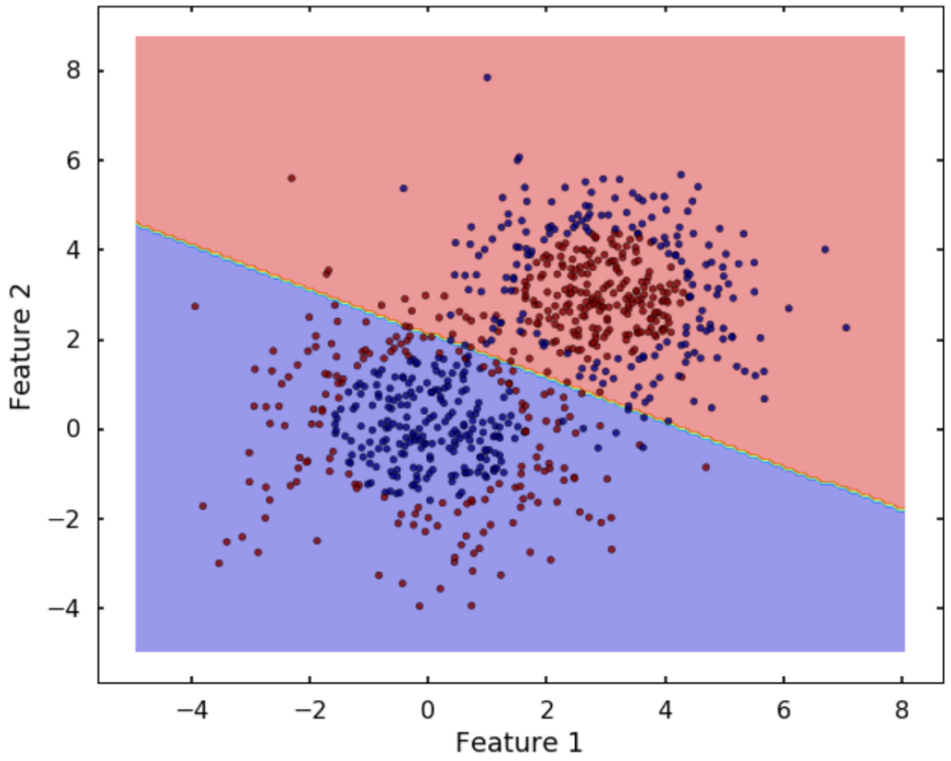
Python example



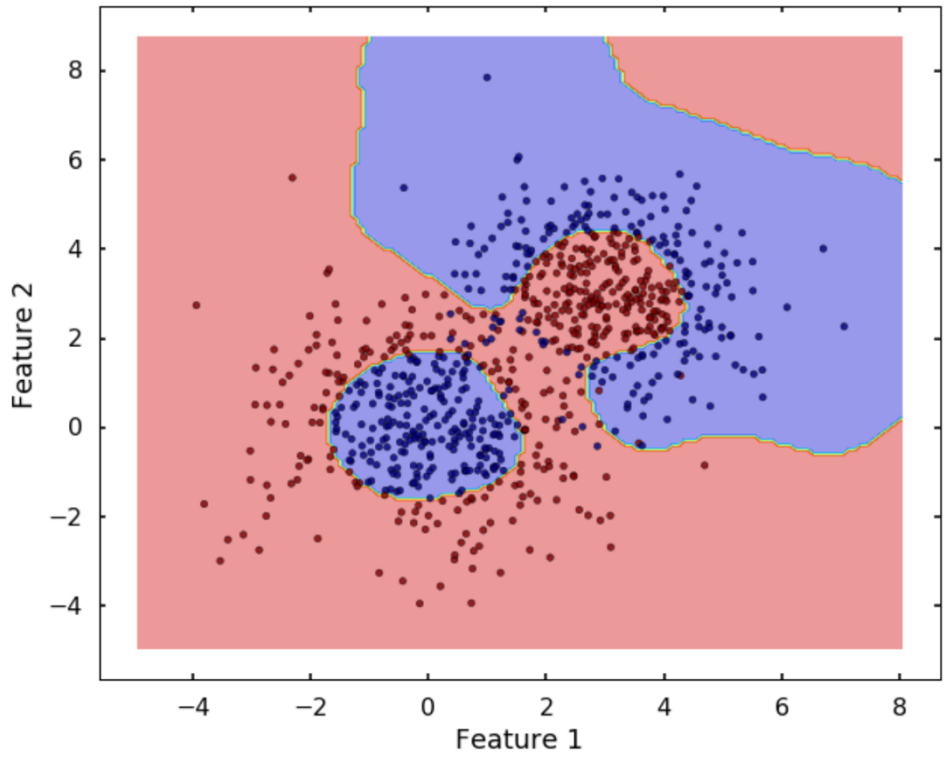
http://qingkaikong.blogspot.com/2016/12/machine-learning-8-support-vector.html

Python example

```
clf = svm.SVC(kernel='linear')
clf.fit(X,y)
```



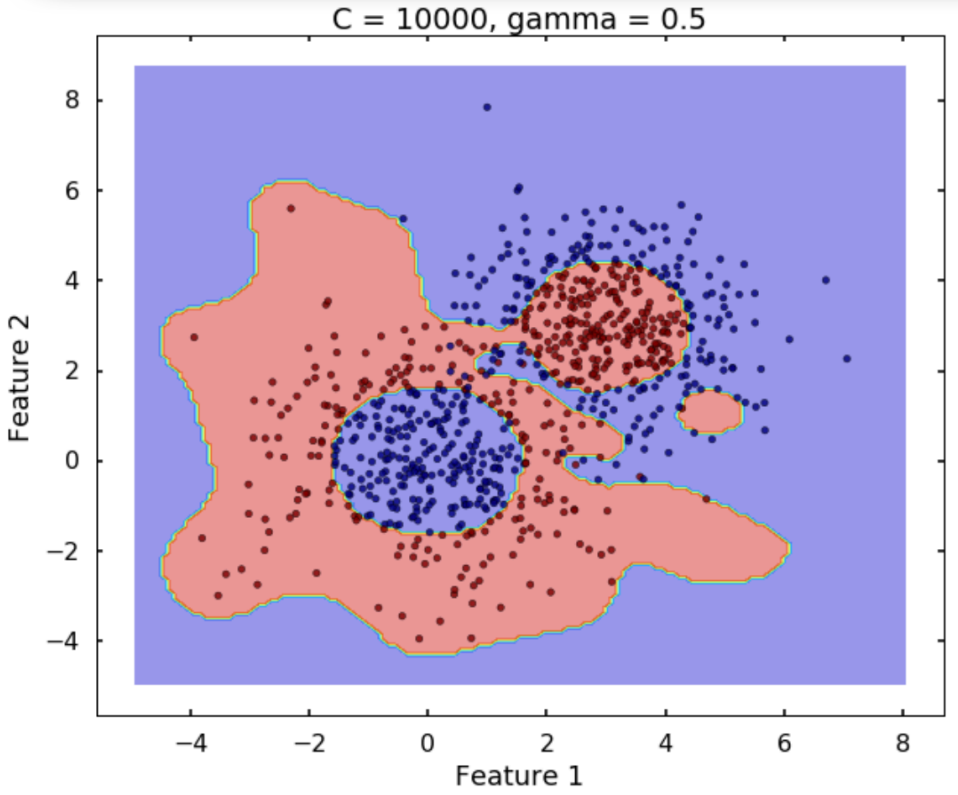

```
clf = svm.SVC(kernel='rbf')
clf.fit(X,y)
```



Python example

- A large C makes the cost of misclassification high.
- This will force the algorithm to fit the data with more flexible model.
- But this will also make the SVM more easier to overit the data.

```
clf = svm.SVC(kernel='rbf', C = 10000, gamma = 0.5)
clf.fit(X,y)
```



Machine Learning Pipeline

Machine Learning Pipeline



EDUCBA



DATA COLLECTION

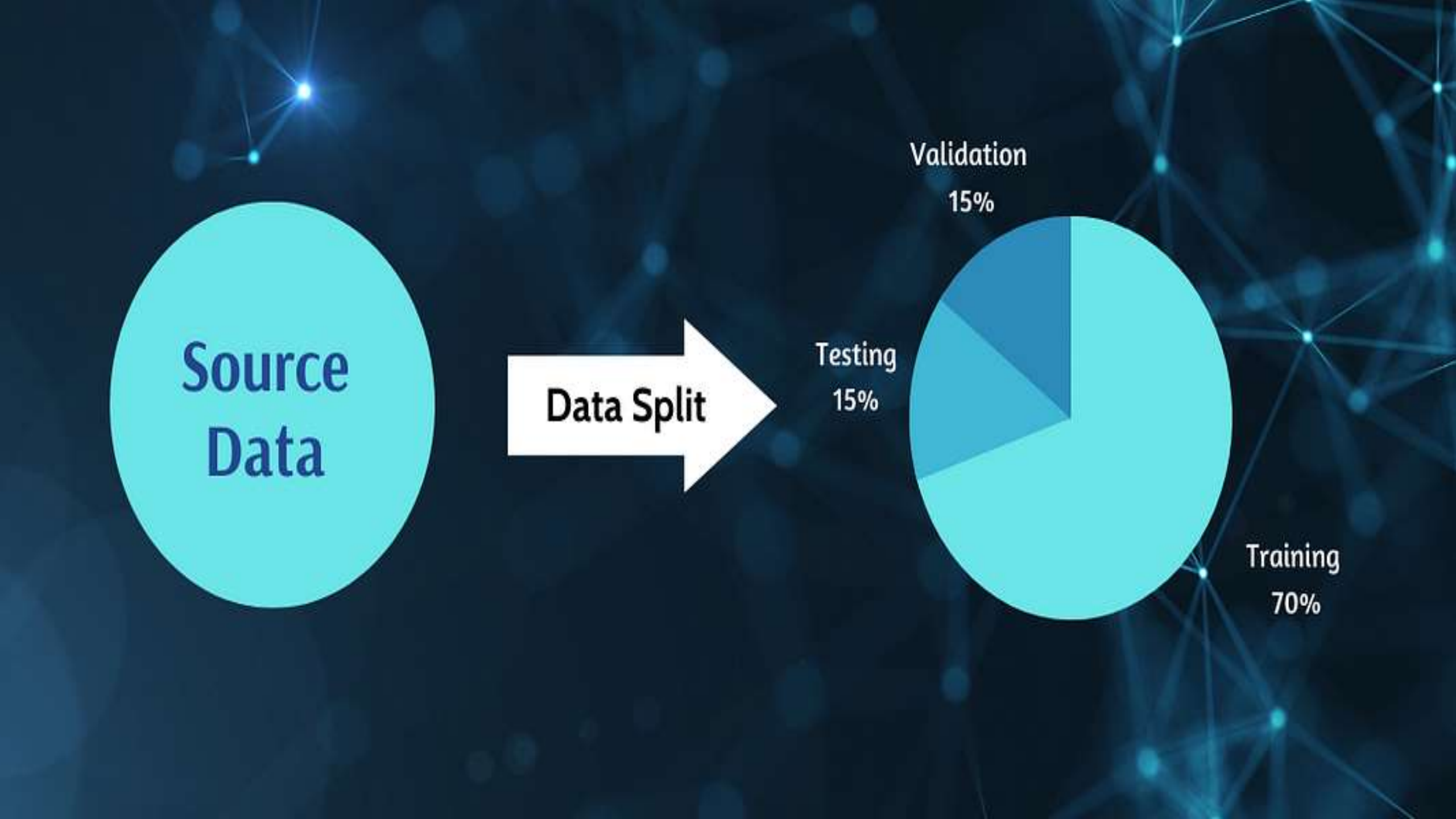
Data Collection

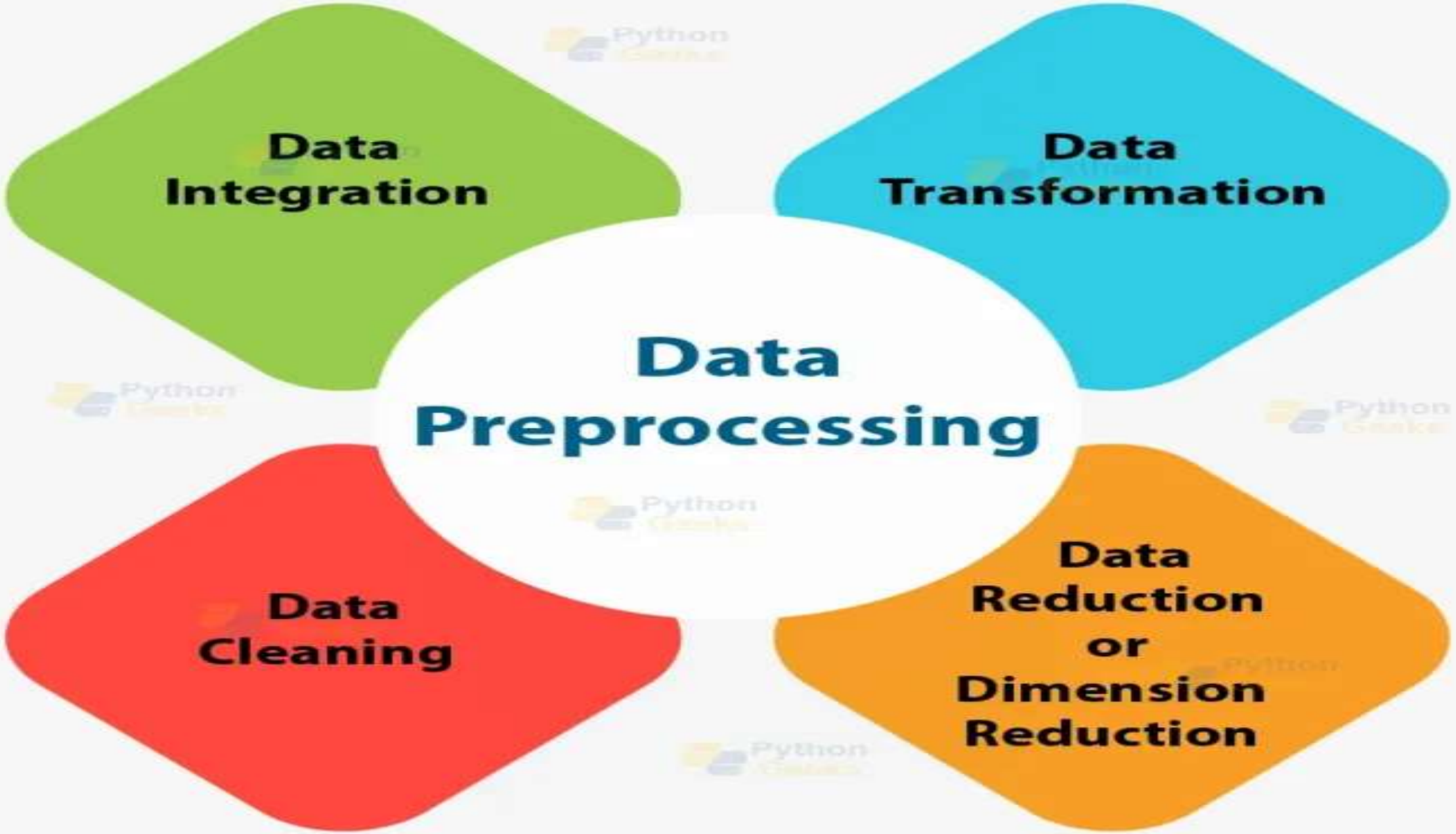
Technical Explanation:

Data collection is the process of gathering raw data from various sources.

This data can be structured (like databases) or unstructured (like text, images, or videos).

Data can come from APIs, web scraping, sensors, or manual entry.


Source Data
Data Split
Validation 15%
Testing 15%
Training 70%


Data Integration
Data Transformation
Data Preprocessing
Data Cleaning
Data Reduction or Dimension Reduction

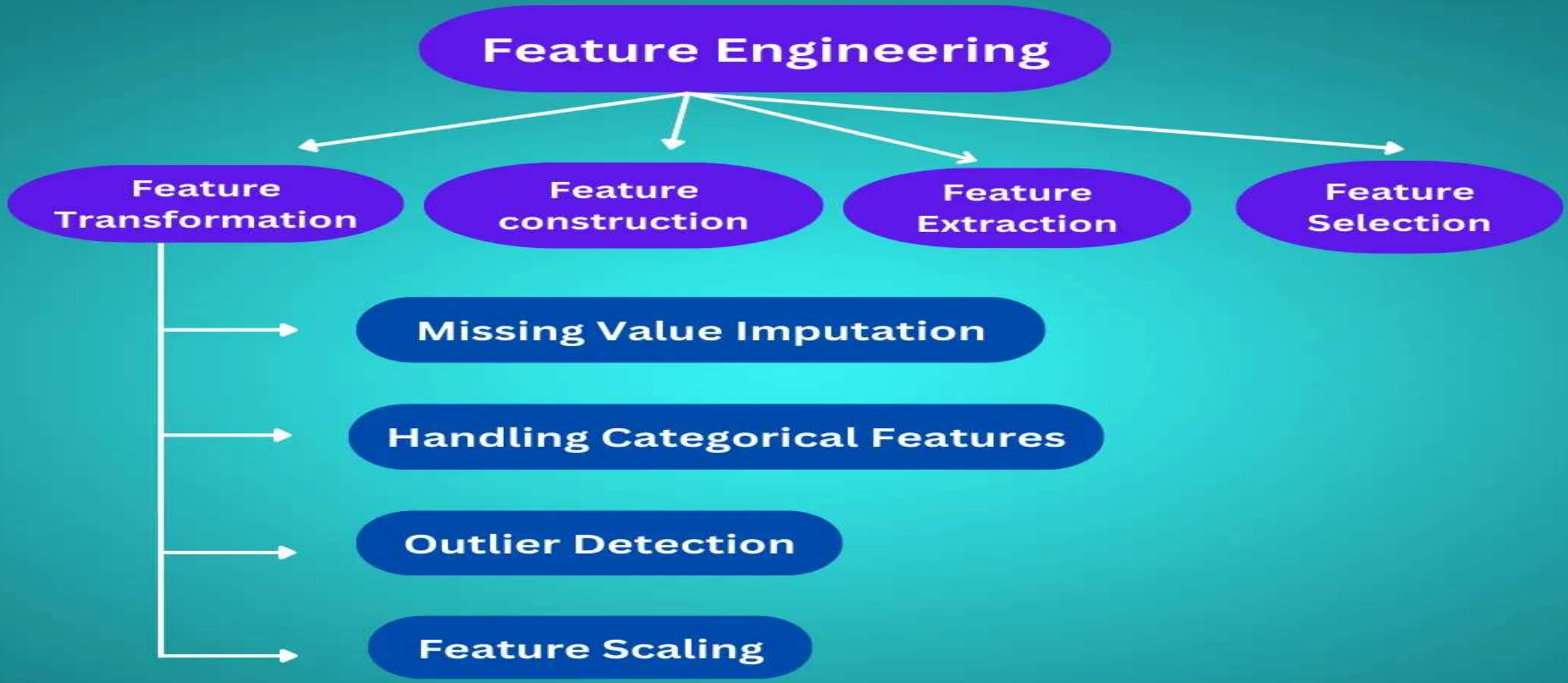
Data Preprocessing

Technical Explanation:

Data preprocessing involves cleaning and transforming the raw data into a format suitable for analysis.

This step includes handling missing values, normalizing data, encoding categorical variables, and splitting the data into training and testing sets.

@uttam-kumar



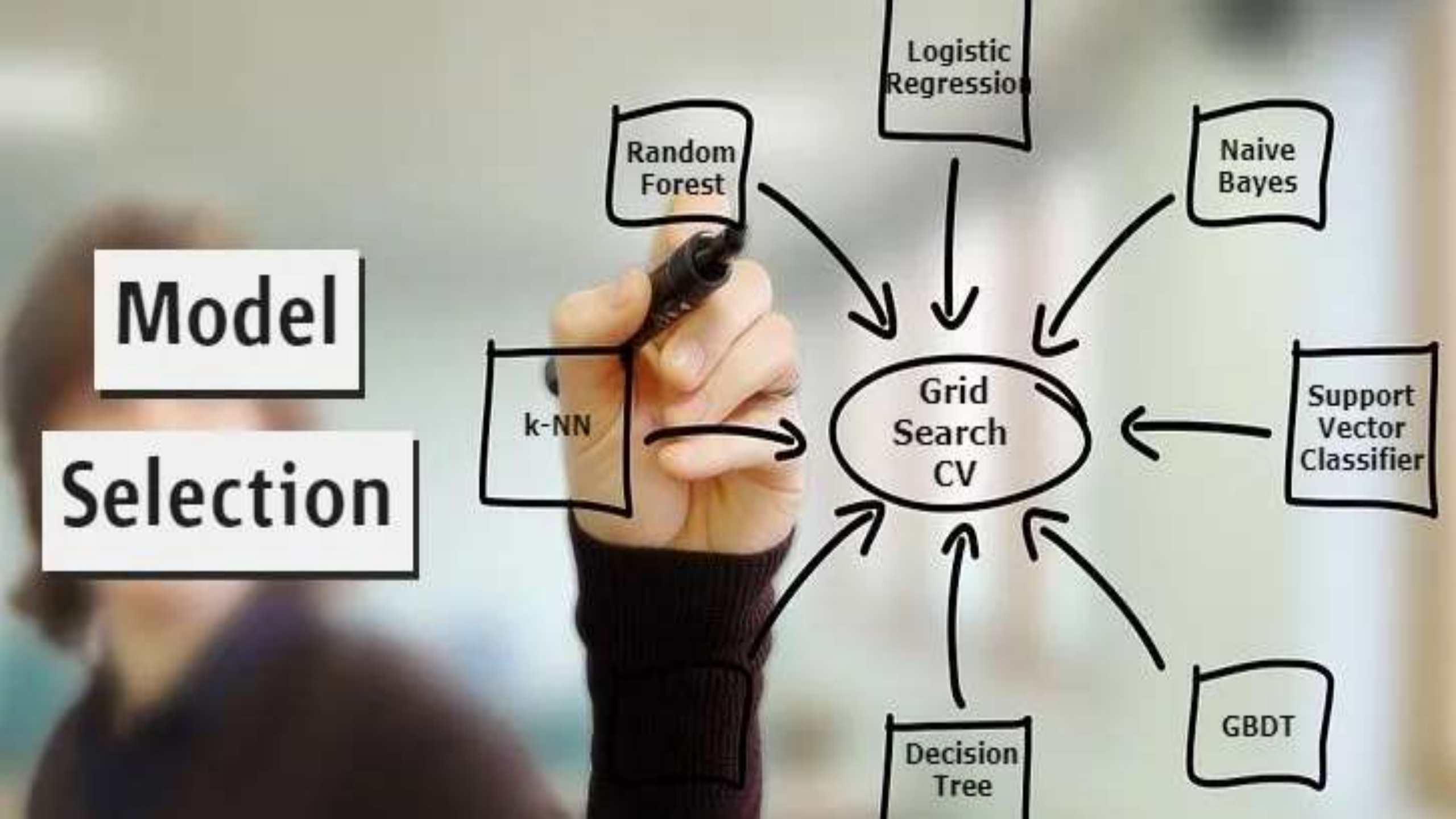
@uttamkumar15802

Feature Engineering

Technical Explanation:

Feature engineering is the process of selecting, modifying, or creating new features from the raw data that can improve the performance of the ML model.

This step often involves domain knowledge and creativity.


Model
Selection
Logistic Regression
Random Forest
Naive Bayes
k-NN
Grid Search CV
Support Vector Classifier
Decision Tree
GBDT

Model Selection

Technical Explanation:

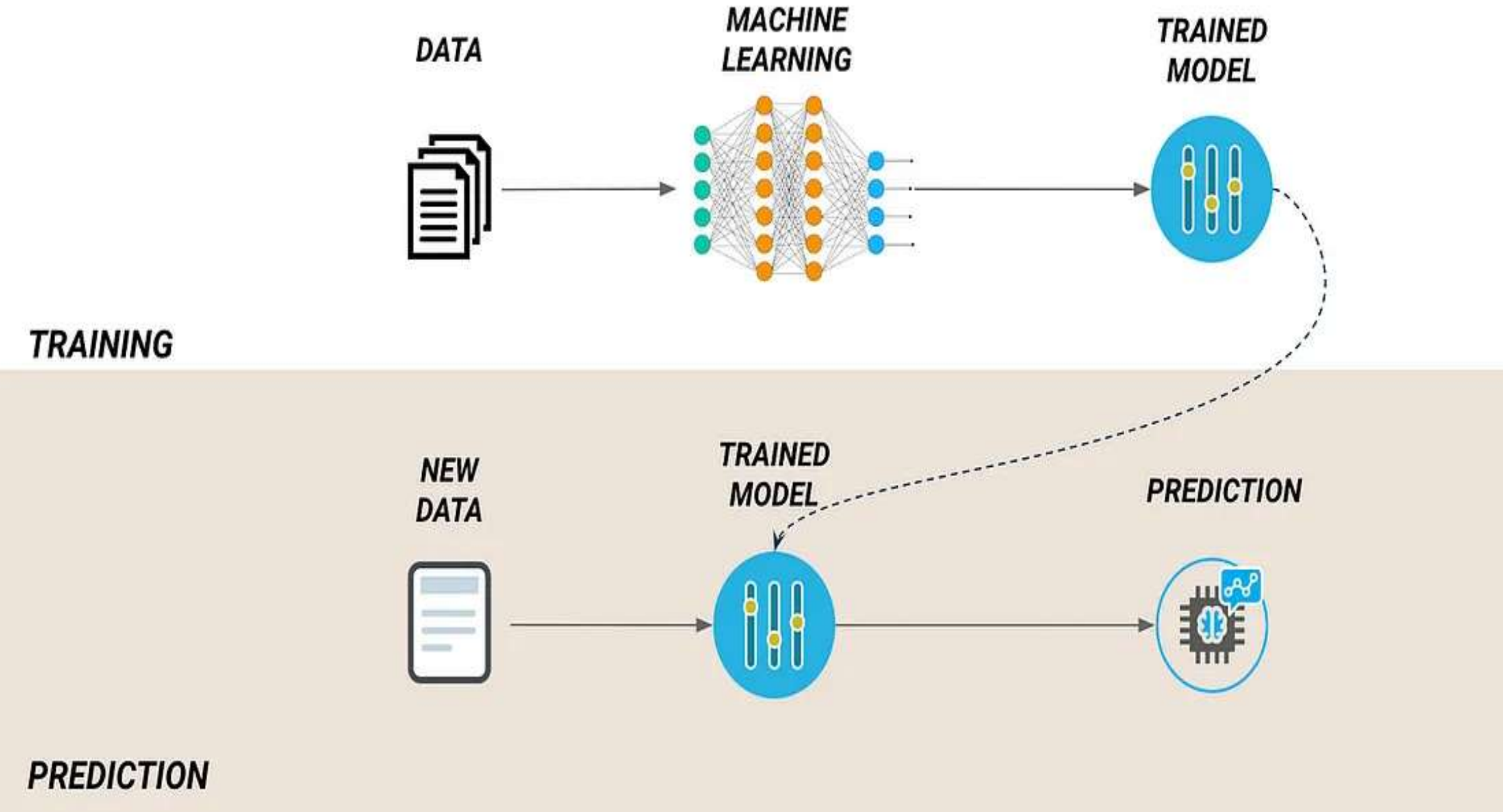
Model selection involves choosing the appropriate machine learning algorithm to train on your data.

Common algorithms include linear regression, decision trees, support vector machines, and neural networks.

The choice depends on the problem type (classification or regression), data size, and complexity.



Model Training in Machine Learning

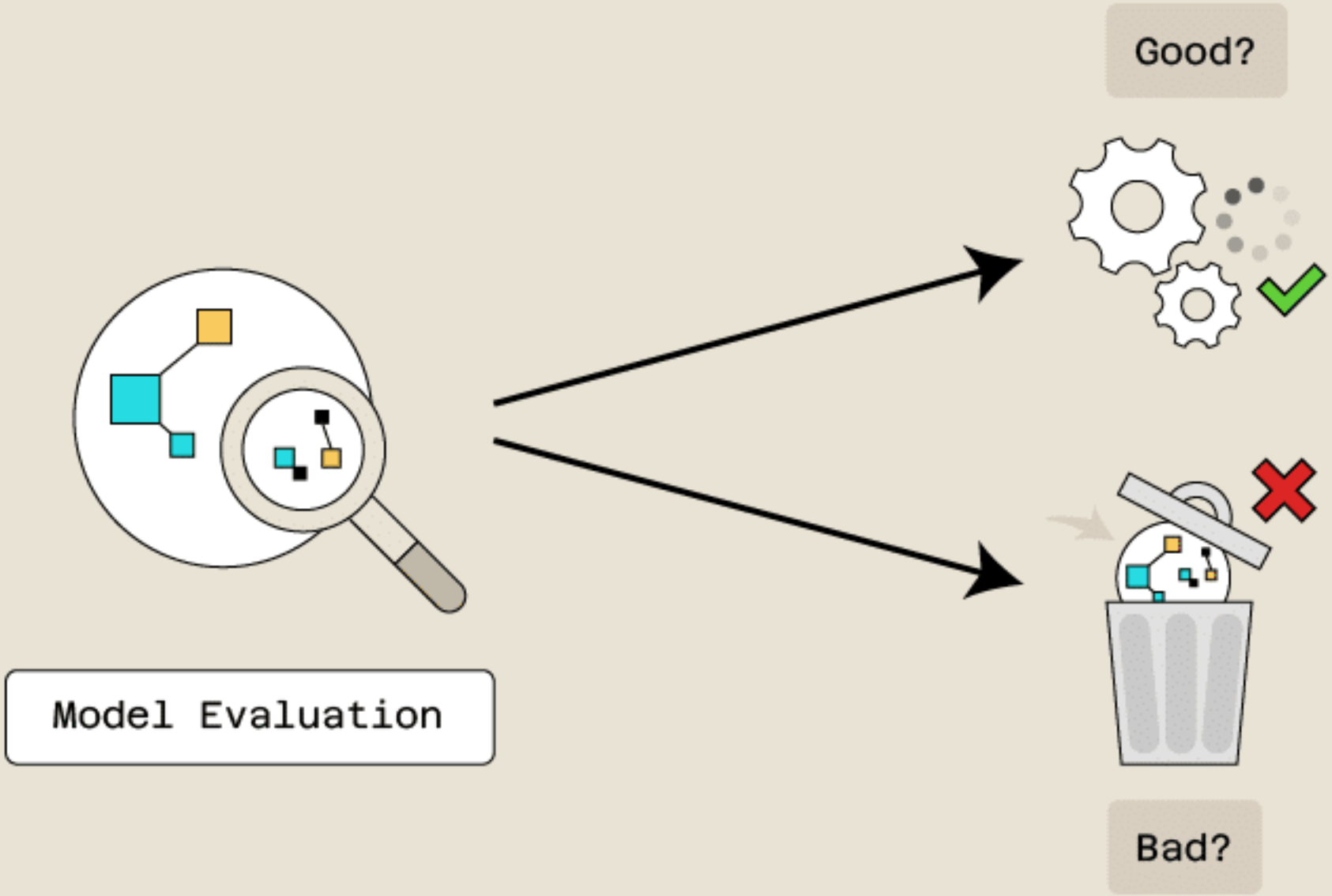

DATA
MACHINE LEARNING
TRAINED MODEL
TRAINING
NEW DATA
TRAINED MODEL
PREDICTION
PREDICTION

Model Training

Technical Explanation:

Model training is the process of feeding the prepared data into the selected algorithm to learn patterns and relationships.

The model adjusts its parameters to minimize error and improve prediction accuracy.


Good?
Model Evaluation
Bad?

Model Evaluation

Technical Explanation:

Model evaluation involves assessing the trained model’s performance using the testing dataset.

Common evaluation metrics include accuracy, precision, recall, F1 score, and ROC-AUC.

MACHINE LEARNING MODEL DEPLOYMENT

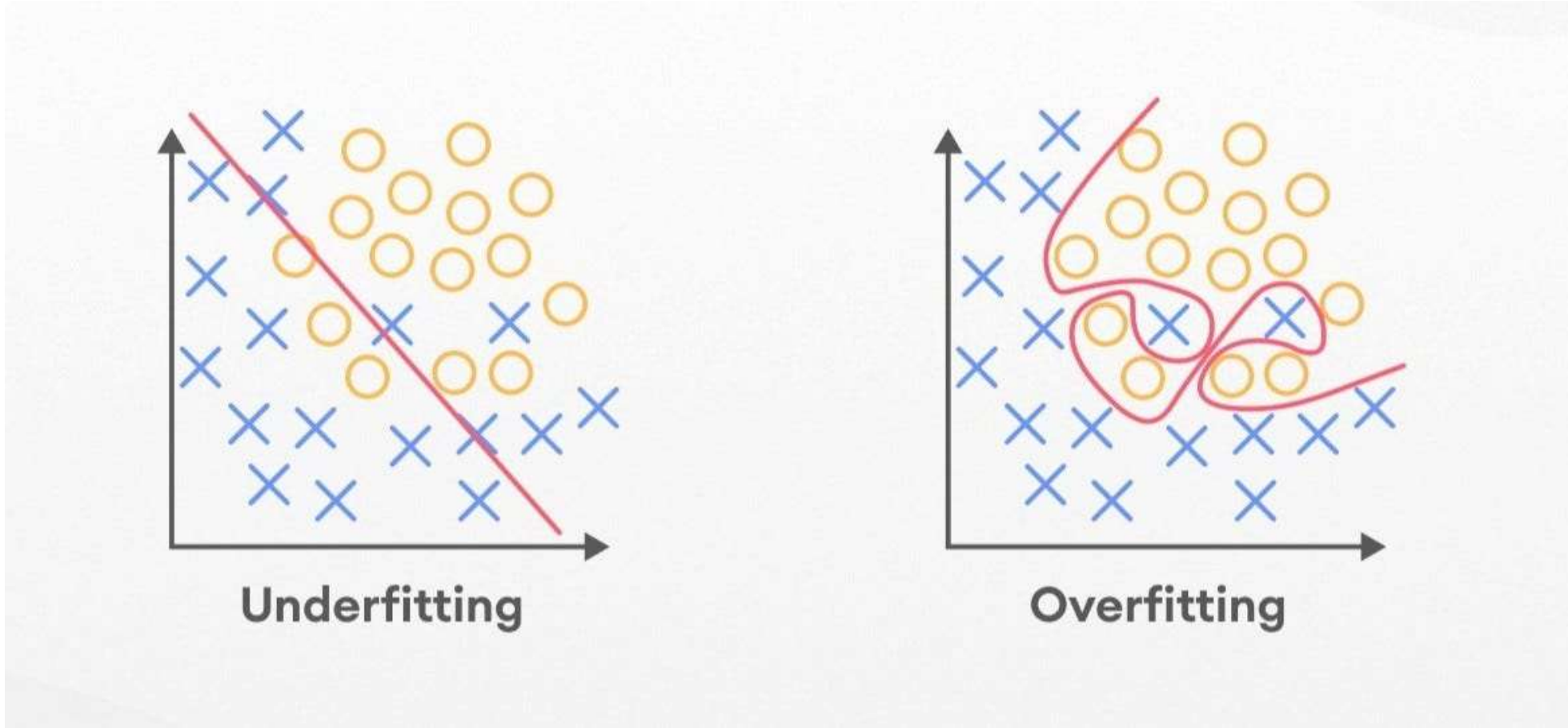


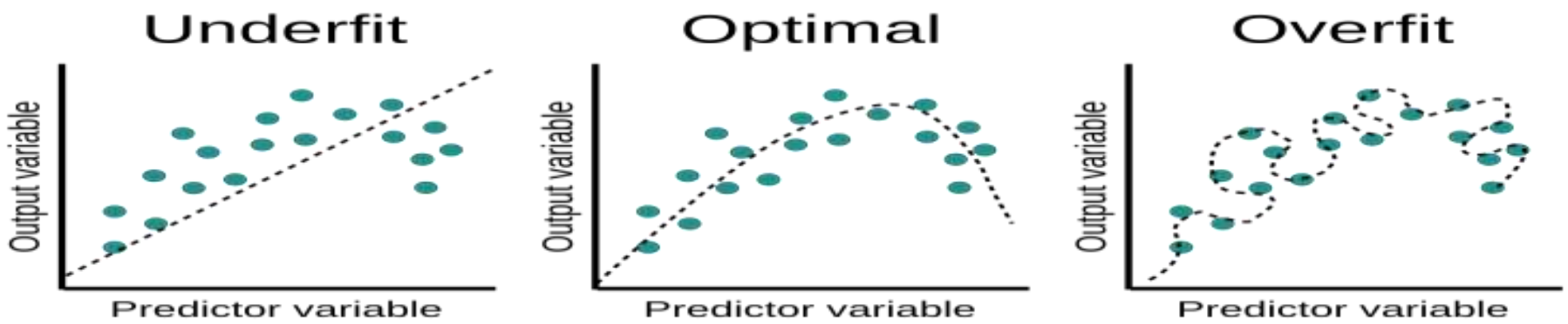
Model Deployment

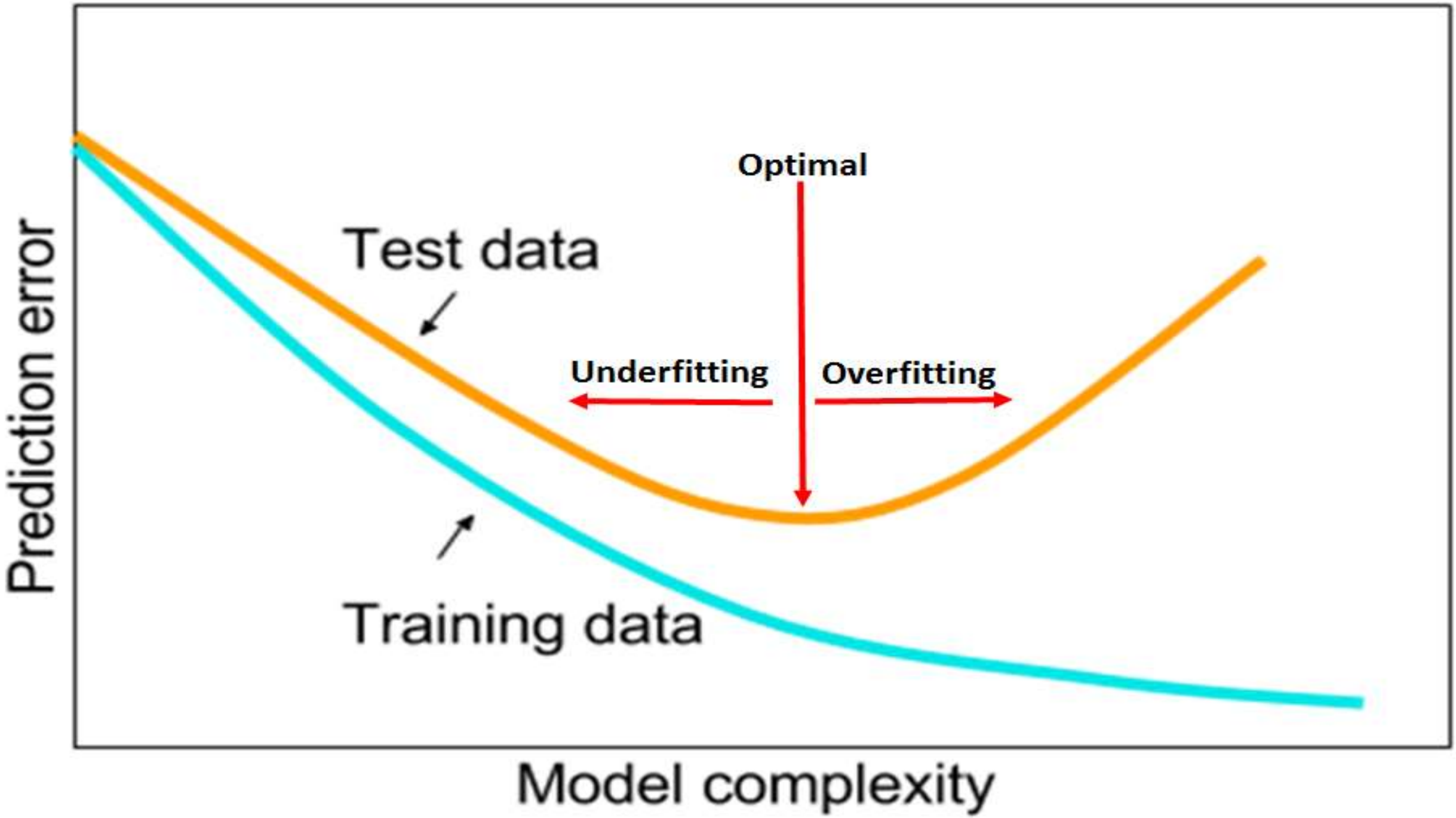
Technical Explanation:

Model deployment is the process of integrating the trained ML model into a production environment where it can make predictions on new data.

This often involves setting up APIs, integrating with web or mobile applications, and ensuring scalability and reliability.


Underfitting
Overfitting


Underfit
Optimal
Overfit
Output variable
Predictor variable
Output variable
Predictor variable
Output variable
Predictor variable


Optimal
Test data
Underfitting
Overfitting
Training data
Prediction error
Model complexity

References:

A)

Course Title : Neural Networks with Tensorflow

Instructor : Sanasam Ranbir Singh
Teaching Assistant: Thiyam Jennil
Lesson 2: Introduction to Machine Learning

What is Machine Learning?(

My First Machine Learning Model??

https://www.iitg.ac.in/cseweb/osint/iNN-week-1.php

B)

Machine Learning Pipeline: From Raw Data to Insightful Models

Medium/Suraj Agrahari

https://medium.com/@surajagrahari330/machine-learning-pipeline-from-raw-data-to-insightful-models-80c54914b7a9

C) Data Mining Slides_Amirkhani_Qom university